CHOIR

A Fully On-Device Neural Voice Synthesis Engine
for the Apple Ecosystem

SOFTWARE REQUIREMENTS SPECIFICATION

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"CHOIR" is a working title and may be renamed without impact on any requirement herein.

Table of Contents

Part I — Foundations

1. Introduction, Purpose & Vision

1.1 Purpose of this Document

This Software Requirements Specification (SRS) defines, exhaustively and testably, everything that the CHOIR voice synthesis package must accomplish. It is the single authoritative source of truth for the project. Every architectural decision, model-training choice, API signature, and platform behaviour built later must trace back to a numbered requirement in this document. Where this document is silent, the developer decides; where this document speaks, it governs.

1.2 Product Vision

CHOIR is a **completely self-contained, from-scratch, on-device neural text-to-speech engine** delivered as a Swift Package for the entire Apple ecosystem (iOS, iPadOS, macOS, visionOS, watchOS, and optionally tvOS). It ships with **32 original, designed synthetic voices** — voices that belong to no real human being, engineered from acoustic first principles to be beautiful, characterful, and production-usable. The library is the permanent, final answer to every text-to-speech need across all of the developer's current and future projects: Scripture reading in THE ONE, narration of translated documents in Ponte, full character voicing for games, and voice-over for video production. Once CHOIR exists, no external TTS service (ElevenLabs, Azure, Google, OpenAI, Apple's server-side voices) is ever required again.

1.3 Guiding Principles

Five principles shape every requirement in this document:

1. **Sovereignty.** Zero network dependency at runtime. No API keys, no rate limits, no per-character billing, no telemetry. The engine must function identically in aeroplane mode on a mountaintop.
2. **Beauty over mimicry.** The voices are not clones of real speakers and do not chase photorealistic human deception. They are *designed instruments* — honest about being synthetic, beautiful because every acoustic parameter was chosen deliberately.
3. **Completeness.** The 32-voice library plus its parametric controls must cover the practical character space of games, narration, audiobooks, and app UI speech, so that no project ever hits a "we don't have a voice for this" wall.
4. **Craftsmanship.** Professional-grade audio (no buzz, no metallic ringing, no phase smearing), a Swift API that feels native and obvious, and documentation good enough that the developer's future self, two years from now, can integrate it in ten minutes.
5. **Permanence.** The package must be maintainable by a single developer indefinitely: reproducible training pipeline, pinned toolchains, no exotic dependencies, no licences that could ever poison a commercial App Store product.

1.4 Intended Audience

The sole developer (author, architect, ML engineer, and consumer), and any AI development assistant working under her direction. The document is written so that any competent engineer could implement the system from it without further clarification.

2. Scope, Non-Goals & Product Boundaries

2.1 In Scope

- A Swift Package (`CHOIR`) containing the runtime synthesis engine, all voice model assets (or an asset-delivery mechanism for them), and the public API.
- 32 fully specified synthetic voices spanning four age bands × two gender presentations, including six villain archetypes.
- A complete linguistic front end for English text: normalization, grapheme-to-phoneme conversion, stress assignment, phrasing, and SSML-like markup control.
- A neural acoustic model and neural vocoder, trained from scratch or from designed/synthetic data, converted to Core ML and executed on-device.
- Parametric runtime control: pitch, rate, emotional intensity, breathiness, age-shift and gender-shift modifiers, pauses, and emphasis.
- Streaming (incremental) synthesis and batch (offline render) synthesis.
- Audio export to standard formats; integration hooks for AVFoundation, spatial audio, games, and timeline-based video workflows.
- A companion showcase/reference application (specified only where it constrains the package; the app has its own future design document).

2.2 Explicit Non-Goals

The following are permanently out of scope. Requirements must never be interpreted as implying them.

- **Voice cloning of real people.** The engine shall provide no capability to imitate a specific real human's voice, and no training pathway that ingests a target speaker's recordings for cloning purposes.
- **Cross-platform runtimes.** No Android, Windows, Linux, or web browser runtime targets. (The *training* pipeline may run on any OS; the *runtime* is Apple-only.)
- **Server-side synthesis.** No hosted inference component of any kind.
- **Singing synthesis** in v1.0. Pitch-contour musicality within speech is in scope; melodic singing to a score is deferred to a future major version.
- **Speech recognition / transcription.** CHOIR is output-only.
- **Real-time voice conversion** (changing a live microphone voice into a CHOIR voice).

2.3 Product Boundary Diagram (textual)

Inside the boundary: text/markup input → normalization → phonemization → prosody prediction → acoustic model → vocoder → PCM audio → playback/export/streaming interfaces → cache. Outside the boundary: the consuming app's UI, its audio session policy decisions (though CHOIR must cooperate with them), the App Store distribution of consuming apps, and the offline training pipeline (specified in Part VI only to the extent that it constrains runtime deliverables).

3. Document Conventions & Requirement Grammar

3.1 Requirement Keywords

Requirement language follows RFC 2119 / RFC 8174 conventions:

MUST / SHALL MUST — an absolute requirement; the product is non-conformant without it.
SHOULD SHOULD — expected in v1.0 unless a documented engineering reason justifies deferral.
MAY MAY — optional; valuable but sacrificeable without recording a deviation.

3.2 Requirement Identifiers

Every requirement carries a stable ID of the form `AREA-NNN` (e.g., `SYN-014`). IDs are never reused or renumbered after publication; withdrawn requirements are marked *Withdrawn* in future revisions rather than deleted. Sub-requirements use letters (`SYN-014a`).

3.3 Requirement Anatomy

Each requirement block contains: the ID, a priority chip, the normative statement, and — where useful — a *Rationale* (italic grey) and *Acceptance* criteria (the objective test by which the requirement is judged met). Acceptance criteria are the seed of the future test plan.

3.4 Numbers and Units

Latency figures are wall-clock milliseconds on the named reference devices (§28.1). Audio measurements assume 48 kHz sample rate unless stated. Frequencies in Hz; loudness in LUFS (ITU-R BS.1770-4); real-time factor (RTF) = synthesis time ÷ audio duration, lower is better.

4. Definitions, Acronyms & Glossary

Term	Definition
Acoustic model	Neural network mapping phoneme/prosody features to mel-spectrogram frames (and predicting duration, F0, and energy).
Vocoder	Neural network mapping mel-spectrogram frames to a raw PCM waveform.
Mel-spectrogram	Time–frequency representation on the mel scale; the interchange format between acoustic model and vocoder.
F0	Fundamental frequency — perceived pitch of the voice.
Formant	Resonant frequency band of the vocal tract (F1–F4); formant structure largely determines perceived vowel identity, age, and gender.
Prosody	Suprasegmental features: pitch contour, rhythm, tempo, pausing, loudness, and phrasing.

G2P	Grapheme-to-phoneme conversion — turning written words into phoneme sequences.
Voice profile	The complete parameter bundle defining one of the 32 voices: conditioning embedding, base F0 range, formant shift, tempo, breathiness, spectral tilt, prosodic style, and metadata.
Voice embedding	The learned conditioning vector selecting a voice inside the shared acoustic model.
RTF	Real-time factor: synthesis wall time divided by output audio duration.
TTFA	Time to first audio: latency from synthesis request to first audible sample in streaming mode.
SSML-C	"SSML for CHOIR" — the package's inline markup dialect for pauses, emphasis, pitch, rate, and phoneme overrides (§10.5).
ANE	Apple Neural Engine — the dedicated ML accelerator on A-series and M-series silicon.
Core ML	Apple's on-device model execution framework; the mandatory deployment format for all CHOIR networks.
Age band	One of: Child, Young Adult (twenties), Middle-Aged, Elderly.
Villain voice	A voice profile deliberately engineered for menace/unease within its age band, per §9.
Consuming app	Any application importing the CHOIR package: THE ONE, Ponte, games, video tools, the showcase app, or future projects.
MOS	Mean Opinion Score — 1–5 subjective audio quality rating (here, self-administered structured listening tests, §29).

5. Stakeholders, Users & Consuming Applications

5.1 Stakeholder

A single stakeholder — the developer — plays every role: product owner, engineer, and primary end user. Secondary end users are the eventual users of consuming apps (readers of THE ONE, players of future games, viewers of produced video), who experience CHOIR only through its audio output. Their needs are represented by the quality requirements of Part VII.

5.2 Known Consuming Applications (v1.0 planning set)

Application	Primary CHOIR usage	Dominant mode	Key sections
THE ONE (Bible study app)	Read-aloud of Scripture and corpus works; verse-synchronised narration	Batch render + cached playback	§33.1, §14, §15
Ponte (PDF translation)	Audio rendering of translated documents; long-document audiobook export	Long-form batch export	§33.2, §14.6
Games (future)	NPC dialogue, cutscenes, dynamic lines	Streaming, low-latency	§33.3, §13

Video production tools	Narration & character voice-over with timeline timing data	Batch render + timing metadata	§33.4, §14.7
Showcase app	Voice gallery, synthesizer UI, export	All modes	§33.5

> **Design consequence.** Because consuming apps span latency-critical streaming (games) and quality-critical long-form rendering (audiobooks), the engine exposes two synthesis paths with a shared model stack (§11, §13) rather than optimizing for a single mode.

Part II — The Voice Library

6. Voice Library General Requirements (VOX-G)

VOX-G-001 MUST The package shall ship exactly **32 voice profiles** organized as 4 age bands (Child, Young Adult, Middle-Aged, Elderly) × 2 gender presentations (male, female) × 4 voices per cell, of which exactly 1 voice per cell is a villain archetype.

Acceptance: `Voice.allCases.count == 32`; filtering by any (ageBand, gender) pair yields 4 voices, exactly one with `isVillain == true`.

VOX-G-002 MUST Every voice shall be an original synthetic design. No voice may be trained to imitate, or be marketed as resembling, any identifiable real person, living or dead.

Rationale: ethical sovereignty and legal safety (right-of-publicity, App Store review).

VOX-G-003 MUST Each voice shall be perceptually distinct from every other voice: in a self-administered ABX-style identification test using 10-second neutral passages, the developer shall correctly identify the voice from its audio at ≥ 90% accuracy across all 32 voices.

Rationale: 32 voices are worthless if 12 of them blur together. Distinctness is a requirement, not an aspiration.

VOX-G-004 MUST Each voice shall carry immutable metadata: stable string identifier (e.g., `"choir.elderly.male.villain.grimshaw"`), display name, age band, gender presentation, villain flag, one-line character description, default parameter set, and recommended-use tags (e.g., `narration`, `dialogue`, `ui`, `antagonist`).

VOX-G-005 MUST Voice identifiers shall be permanent across all future versions of the package. New voices may be added; existing identifiers may never change meaning or be removed within a major version.

Rationale: a saved game or cached audiobook must resolve the same voice years later.

VOX-G-006 MUST All 32 voices shall share a single acoustic model and a single vocoder (or at most one vocoder per age-band family), selected via conditioning embeddings — not 32 separate model files.

Rationale: disk footprint, memory ceiling on watchOS/older iPhones, and maintainability.

Acceptance: total on-disk model asset size complies with PRF-030; loading a second voice after a first does not reload the network weights.

VOX-G-007 MUST Every voice shall remain intelligible and in-character across its full documented customization envelope (§12.9): the pitch, rate, emotion, and breathiness ranges published for that voice.

Acceptance: the Harvard-sentence intelligibility test of QUA-004 passes at every corner of the envelope (min/max pitch × min/max rate).

VOX-G-008 SHOULD Each voice shall ship with three built-in demonstration passages (neutral narration, emotive dialogue, and a character-typical line) synthesizable by a single API call, for gallery/preview use.

VOX-G-009 MAY The package may expose a "voice designer" API for deriving custom voices by interpolating between voice embeddings; if provided, derived voices must be serializable and restorable by the consuming app.

VOX-G-010 MUST Child voices shall be stylized child *characters* — bright, small-vocal-tract timbres appropriate for fiction — and shall not be engineered to be mistaken for recordings of real children.

Rationale: same ethical boundary as VOX-G-002, applied specifically to minors; the goal is storybook character, not simulation of an actual child.

7. Acoustic Design Parameters & Age/Gender Modelling

This section defines the shared parameter vocabulary used by every voice profile in §8, and the acoustic theory of how age, gender, and menace are realized. These are design requirements on the training pipeline and conditioning system.

7.1 The Voice Profile Parameter Set

VOX-P-001 MUST Every voice profile shall be fully described by, at minimum, the following designed parameters, each stored in the profile and honoured by the models: median F0 and F0 range (Hz); formant scale factor (vocal-tract length proxy); spectral tilt (dB/octave brightness); breathiness index (0–1); roughness/rasp index (0–1, realized via controlled jitter/shimmer and subharmonic energy); base tempo (syllables/second); pause style (duration multipliers for comma/period/paragraph); pitch dynamism (semitone standard deviation of the intonation contour); and articulation precision (0–1, consonant crispness).

7.2 Age Realization Requirements

VOX-P-002 MUST **Child voices** shall be realized with: high median F0 (male 210–290 Hz, female 230–320 Hz), formant scale ≈ 1.15–1.25 (short vocal tract), low roughness, high articulation energy with age-typical slight softening of /r/ and /s/ crispness *as a stylistic option only*, faster micro-tempo but shorter breath groups, and wide, mobile pitch contours.

VOX-P-003 MUST **Young Adult voices** shall be realized with: male median F0 100–135 Hz, female 190–230 Hz; formant scale ≈ 1.0; low-to-moderate breathiness; clean periodicity (minimal rasp); confident phrase-final movement; tempo 4.2–5.2 syll/s.

VOX-P-004 MUST **Middle-Aged voices** shall be realized with: male median F0 90–120 Hz, female 175–210 Hz; formant scale 0.97–1.0; slightly increased spectral tilt (darker); measured tempo 3.8–4.6 syll/s; authoritative flat-to-falling declarative contours with controlled dynamism.

`VOX-P-005` **MUST** **Elderly voices** shall be realized with: male median F0 95–130 Hz (age-raised), female 160–195 Hz (age-lowered); mild-to-moderate roughness (jitter/shimmer character) and breathiness; tempo 3.2–4.0 syll/s with longer pauses; slightly reduced articulation precision; and gentle F0 instability (±1 semitone slow drift) as a *designed, pleasant* texture — never degradation artifacts.

7.3 Gender Realization Requirements

`VOX-P-006` **MUST** Gender presentation shall be realized primarily through the joint setting of F0 range and formant scale (female ≈ +15–20% formant scale relative to male within the same age band), with breathiness and pitch-dynamism as secondary cues — never through F0 alone.

Rationale: pitch-only "gender" sounds like a chipmunk filter; formant structure is what actually carries it.

7.4 The Runtime Age/Gender Modifiers

`VOX-P-007` **MUST** The engine shall expose continuous runtime modifiers `ageShift ∈ [−1, +1]` and `genderShift ∈ [−1, +1]` which move a voice within (not across) its perceptual neighbourhood by scaling F0, formant scale, tempo, and roughness along the trajectories of §7.2–7.3, with documented per-voice safe ranges.

Acceptance: at ±0.5 on either modifier every voice still passes VOX-G-003 distinctness and QUA-004 intelligibility.

8. The 32 Voice Profiles — Full Specifications

Each profile below is a binding requirement (`VOX-08 …` series, one per voice): the shipped voice must match its specification within the tolerances of §29.6. Names are working names; the character specification, not the name, is normative. Parameter abbreviations: **F0** median (range); **FSc** formant scale; **Tilt** brightness (higher = brighter); **Br** breathiness; **Rg** roughness; **Tempo** syll/s; **Dyn** pitch dynamism.

8.1 Child — Male (4)

FINCH — bright, curious boy `choir.child.male.finch · VOX-08-01`

Character	Eager question-asker; sunlit, quick, open-hearted. The default "young hero's friend."
Parameters	F0 250 Hz (200–340) · FSc 1.22 · Tilt bright · Br 0.15 · Rg 0.05 · Tempo 4.8 · Dyn high (3.5 st)
Prosody signature	Rising terminal contours on ~30% of declaratives (genuine curiosity, not uptalk parody); short breath groups; light energetic onsets.
Recommended use	Games (child NPC/protagonist), children's narration, playful UI.

SCOUT — confident older-child explorer

`choir.child.male.scout` · VOX-08-02

Character	10–12-year-old energy; capable, a little cocky, leader of the treehouse.
Parameters	F0 225 Hz (180–300) · FSc 1.16 · Tilt bright-neutral · Br 0.10 · Rg 0.06 · Tempo 5.0 · Dyn med-high (3.0 st)
Prosody signature	Firm falling declaratives; emphatic stress on action verbs; quick pickup after commas.
Recommended use	Adventure-game protagonists, sporty characters, energetic explainers.

ALDER — serious, mature-for-his-age boy

`choir.child.male.alder` · VOX-08-03

Character	The quiet, watchful child; old soul; speaks less, means more.
Parameters	F0 215 Hz (185–265) · FSc 1.15 · Tilt neutral · Br 0.20 · Rg 0.05 · Tempo 4.0 · Dyn low-med (2.0 st)
Prosody signature	Even, deliberate pacing; longer pre-pausal lengthening; narrow but precise pitch movement.
Recommended use	Dramatic child roles, poignant narration, mystery.

WICK — sly child villain

`choir.child.male.villain.wick` · VOX-08-04

Character	Precocious menace; the smiling child who is three moves ahead. Unsettling through control, not growling.
Parameters	F0 235 Hz (195–290) · FSc 1.18 · Tilt neutral-dark for age · Br 0.25 · Rg 0.08 · Tempo 3.9 (unnaturally measured for a child) · Dyn low (1.8 st) with sudden 4-st spikes
Prosody signature	Over-precise articulation; flattened contours with sing-song lapses; micro-pauses before key words (§9 rules apply).
Recommended use	Horror/thriller child antagonists, creepy fairy-tale roles.

8.2 Child — Female (4)

WREN — playful, sparkling girl

`choir.child.female.wren` · VOX-08-05

Character	Giggle-adjacent brightness; delight as a default state.
Parameters	F0 275 Hz (225–370) · FSc 1.24 · Tilt very bright · Br 0.18 · Rg 0.04 · Tempo 5.0 · Dyn high (3.8 st)
Prosody signature	Bouncing stress rhythm; wide rise-falls on exclamatives; short, light pauses.
Recommended use	Children's content, cheerful game companions, playful UI.

JUNIPER — curious storyteller girl `choir.child.female.juniper · VOX-08-06`

Character	The child who narrates her own life; wonder with structure.
Parameters	F0 260 Hz (215–340) · FSc 1.20 · Tilt bright · Br 0.15 · Rg 0.05 · Tempo 4.5 · Dyn med-high (3.2 st)
Prosody signature	Narrative arcs: pitch reset at sentence starts, suspenseful pre-final lengthening ("and *then*...").
Recommended use	Child narrator roles, audiobook children's chapters, quest-giver children.

CLOVER — gentle, innocent girl `choir.child.female.clover · VOX-08-07`

Character	Soft-spoken sweetness; the trusting youngest sibling.
Parameters	F0 265 Hz (225–330) · FSc 1.22 · Tilt bright-soft · Br 0.30 · Rg 0.03 · Tempo 4.1 · Dyn medium (2.6 st)
Prosody signature	Gentle onsets; slightly breathy phrase endings; small rises seeking reassurance.
Recommended use	Tender scenes, lullaby-adjacent narration, vulnerable characters.

BRIAR — unsettlingly charming girl villain `choir.child.female.villain.briar · VOX-08-08`

Character	Porcelain-doll sweetness with something wrong underneath; manipulation wearing innocence.
Parameters	F0 268 Hz (225–345) · FSc 1.21 · Tilt bright with hollowed mids · Br 0.35 · Rg 0.05 · Tempo 3.8 · Dyn low (1.6 st) with sing-song intrusions
Prosody signature	Too-even melody; sweetness sustained past its natural duration; whispered-adjacent intensity on threats.
Recommended use	Horror, dark fantasy, unsettling fairy tales.

8.3 Young Adult — Male (4)

ORION — warm, hopeful young hero `choir.ya.male.orion · VOX-08-09`

Character	Early-twenties idealist; open, earnest, easy to root for.
Parameters	F0 120 Hz (85–190) · FSc 1.0 · Tilt neutral-warm · Br 0.15 · Rg 0.04 · Tempo 4.7 · Dyn med-high (3.0 st)
Prosody signature	Forward momentum; rising pre-nuclear accents; sincere emphasis without theatricality.
Recommended use	Protagonists, romance leads, motivational narration.

FLINT — grounded, capable young man `choir.ya.male.flint · VOX-08-10`

Character	Late-twenties competence; steady hands, dry humour available.
Parameters	F0 108 Hz (80–165) · FSc 0.99 · Tilt neutral-dark · Br 0.10 · Rg 0.07 · Tempo 4.3 · Dyn medium (2.4 st)
Prosody signature	Economical contours; clipped confident finals; understated emphasis.
Recommended use	Action heroes, soldiers/engineers, documentary narration.

REED — soft, vulnerable young man `choir.ya.male.reed · VOX-08-11`

Character	Sensitive interiority; the poet, the grieving brother, the anxious friend.
Parameters	F0 125 Hz (95–180) · FSc 1.01 · Tilt soft · Br 0.35 · Rg 0.04 · Tempo 3.9 · Dyn medium (2.6 st)
Prosody signature	Breathy onsets; hesitation-shaped pausing (without disfluencies); falling-then-fading finals.
Recommended use	Introspective narration, poetry, emotional dialogue.

CORVIN — charismatic predator `choir.ya.male.villain.corvin · VOX-08-12`

Character	The charming young villain; velvet over wire; seduction as strategy.
Parameters	F0 105 Hz (78–160) · FSc 0.98 · Tilt dark-smooth · Br 0.28 · Rg 0.06 · Tempo 3.7 · Dyn low-med (2.0 st) with silken glides
Prosody signature	Slow controlled glides; intimate low-energy delivery that never rushes; smiling-voice resonance on threats.
Recommended use	Romantic-thriller antagonists, cult leaders, con men.

8.4 Young Adult — Female (4)

LYRA — vibrant, determined young woman `choir.ya.female.lyra · VOX-08-13`

Character	Bright drive; warmth plus will; the heroine who moves the plot.
Parameters	F0 210 Hz (160–300) · FSc 1.18 · Tilt bright · Br 0.15 · Rg 0.04 · Tempo 4.8 · Dyn high (3.4 st)
Prosody signature	Energised nuclear accents; decisive falls; quick, clean phrase turnover.
Recommended use	Protagonists, upbeat narration, presenters.

ISLA — grounded, self-assured young woman `choir.ya.female.isla · VOX-08-14`

Character	Calm certainty; low-drama competence; trustworthy centre of gravity.
Parameters	F0 195 Hz (150–260) · FSc 1.16 · Tilt neutral-warm · Br 0.18 · Rg 0.04 · Tempo 4.2 · Dyn medium (2.4 st)
Prosody signature	Level, assured contours; measured emphasis; unhurried finals.
Recommended use	Narration (long-form), professionals, mentors-in-training.

NOVA — intense, burning young woman `choir.ya.female.nova · VOX-08-15`

Character	Contained fire; conviction at the edge of eruption.
Parameters	F0 205 Hz (155–290) · FSc 1.17 · Tilt bright-hard · Br 0.10 · Rg 0.08 · Tempo 4.6 · Dyn high (3.6 st), compressed then released
Prosody signature	Pressurised evenness broken by sharp emphatic peaks; hard consonant attack under emotion.
Recommended use	Dramatic leads, revolutionaries, high-stakes dialogue.

SABLE — cold, brilliant villainess `choir.ya.female.villain.sable · VOX-08-16`

Character	Intelligence as a weapon; beauty with the warmth removed; every word chosen.
Parameters	F0 188 Hz (145–250) · FSc 1.14 · Tilt dark-glassy · Br 0.22 · Rg 0.05 · Tempo 3.8 · Dyn low (1.7 st) with surgical accents
Prosody signature	Near-monotone control; icy micro-pauses; precise consonants; amusement leaked in tiny pitch curls.
Recommended use	Masterminds, corporate antagonists, femme-fatale roles.

8.5 Middle-Aged — Male (4)

GARRICK — authoritative, life-worn leader `choir.mid.male.garrick · VOX-08-17`

Character	Command earned by experience; gravity without cruelty.
Parameters	F0 100 Hz (72–150) · FSc 0.98 · Tilt dark · Br 0.10 · Rg 0.12 · Tempo 4.0 · Dyn medium (2.2 st)
Prosody signature	Weighted stress; long authoritative falls; strategic silence before verdicts.
Recommended use	Kings/commanders, epic narration, trailers.

HALE — warm mentor `choir.mid.male.hale · VOX-08-18`

Character	The teacher you remember; patience, humour, safety.
Parameters	F0 110 Hz (82–160) · FSc 0.99 · Tilt warm · Br 0.20 · Rg 0.08 · Tempo 4.1 · Dyn med (2.6 st)
Prosody signature	Smiling resonance; inclusive rises inviting the listener along; soft landings.
Recommended use	Educational narration, mentors, devotional reading.

BRAM — weary, world-worn man `choir.mid.male.bram · VOX-08-19`

Character	Noir tiredness; decency that has paid its costs.
Parameters	F0 96 Hz (70–140) · FSc 0.98 · Tilt dark-dry · Br 0.28 · Rg 0.16 · Tempo 3.7 · Dyn low-med (2.0 st)
Prosody signature	Heavy phrase-final lowering; sighing exhalation shape; laconic emphasis.
Recommended use	Noir narration, detectives, tragic roles.

MALVERN — calculating, ruthless villain `choir.mid.male.villain.malvern · VOX-08-20`

Character	Institutional evil; the man who signs the order and sleeps well.
Parameters	F0 94 Hz (68–135) · FSc 0.97 · Tilt dark-metallic edge · Br 0.12 · Rg 0.14 · Tempo 3.6 · Dyn very low (1.4 st) with terminal drops
Prosody signature	Flattened affect; unhurried certainty; consonant precision as intimidation; pauses that make the listener wait.
Recommended use	Primary antagonists, tyrants, corporate villains.

8.6 Middle-Aged — Female (4)

MARION — composed, elegant authority `choir.mid.female.marion · VOX-08-21`

Character	Grace with spine; the judge, the abbess, the matriarch in her prime.
Parameters	F0 185 Hz (140–245) · FSc 1.14 · Tilt warm-dark · Br 0.14 · Rg 0.07 · Tempo 3.9 · Dyn medium (2.3 st)
Prosody signature	Poised, legato phrasing; deliberate accent placement; serene authoritative falls.
Recommended use	Authority figures, elegant narration, ceremonial reading.

TAMSIN — poised professional `choir.mid.female.tamsin · VOX-08-22`

Character	Crisp modern competence; broadcast-clean clarity.
Parameters	F0 192 Hz (150–250) · FSc 1.15 · Tilt neutral-bright · Br 0.10 · Rg 0.05 · Tempo 4.4 · Dyn medium (2.5 st)
Prosody signature	Newsreader cadence; clean list intonation; efficient, confident turnover.
Recommended use	App UI speech, documentation reading, presenters.

GREER — dry, weathered woman `choir.mid.female.greer · VOX-08-23`

Character	Sardonic survivor; wit with sandpaper edges.
Parameters	F0 172 Hz (130–225) · FSc 1.12 · Tilt dark-dry · Br 0.24 · Rg 0.15 · Tempo 3.8 · Dyn low-med (2.0 st)
Prosody signature	Deadpan plateaus; late, dry accents; downstepped irony contours.
Recommended use	Hard-bitten roles, dark comedy, war-weary narration.

RAVENNA — sharp, power-hungry villainess `choir.mid.female.villain.ravenna · VOX-08-24`

Character	Ambition with a crown in mind; contempt dressed as courtesy.
Parameters	F0 178 Hz (135–240) · FSc 1.13 · Tilt dark with brilliant edge · Br 0.15 · Rg 0.10 · Tempo 3.7 · Dyn low (1.8 st) with imperious peaks
Prosody signature	Regal downsteps; honeyed openings hardening to steel finals; scorn realised in lengthened stressed vowels.
Recommended use	Evil queens, political antagonists, cold matriarchs.

8.7 Elderly — Male (4)

ALARIC — distinguished, measured elder `choir.eld.male.alaric · VOX-08-25`

Character	The scholar-statesman; every sentence weighed and found exact.
Parameters	F0 112 Hz (85–160) · FSc 0.985 · Tilt dark-mellow · Br 0.22 · Rg 0.18 · Tempo 3.5 · Dyn low-med (2.0 st)
Prosody signature	Stately periodic phrasing; long structural pauses; gravitas without pomposity.
Recommended use	Scholarly narration, prologues, sages.

WILFRED — warm grandfather `choir.eld.male.wilfred · VOX-08-26`

Character	Fireside kindness; stories, chuckles held in reserve, unconditional welcome.
Parameters	F0 118 Hz (90–165) · FSc 0.99 · Tilt warm-soft · Br 0.30 · Rg 0.20 · Tempo 3.4 · Dyn medium (2.4 st)
Prosody signature	Rocking narrative rhythm; affectionate rises; unhurried, comfortable settlings.
Recommended use	Bedtime narration, grandfather roles, gentle devotional reading.

CORMAC — sharp-witted elder `choir.eld.male.cormac · VOX-08-27`

Character	Age without dullness; the old man who still wins every argument.
Parameters	F0 122 Hz (92–175) · FSc 0.99 · Tilt neutral-dry · Br 0.20 · Rg 0.17 · Tempo 3.9 (fast for his band) · Dyn med-high (2.8 st)
Prosody signature	Quick parenthetical asides; puckish accent placement; crisp comic timing in pause structure.
Recommended use	Comic elders, wizards with attitude, lively memoir narration.

GRIMSHAW — gravelly, menacing elder villain

Character	Accumulated darkness; a lifetime of cruelty distilled into patience.
Parameters	F0 98 Hz (70–140) · FSc 0.98 · Tilt very dark, subharmonic body · Br 0.25 · Rg 0.30 (controlled gravel) · Tempo 3.1 · Dyn low (1.5 st) with cavernous terminal falls
Prosody signature	Geological pacing; threats delivered as observations; rasp intensity rises with intimacy, not volume.
Recommended use	Dark lords, ancient evils, horror narrators.

8.8 Elderly — Female (4)

MAEVE — nurturing wise grandmother `choir.eld.female.maeve · VOX-08-29`

Character	Maternal wisdom; the blessing voice; tea, scripture, and certainty that you are loved.
Parameters	F0 175 Hz (135–225) · FSc 1.11 · Tilt warm-soft · Br 0.32 · Rg 0.15 · Tempo 3.4 · Dyn medium (2.3 st)
Prosody signature	Cradling contours; benedictory finals; patient, spacious pausing.
Recommended use	Comforting narration, grandmothers, Scripture reading.

ODETTE — serene elder `choir.eld.female.odette · VOX-08-30`

Character	Stillness after the storm; contemplative depth; peace that was fought for.
Parameters	F0 168 Hz (130–210) · FSc 1.10 · Tilt dark-luminous · Br 0.28 · Rg 0.12 · Tempo 3.2 · Dyn low (1.8 st)
Prosody signature	Long even arcs; minimal but exact emphasis; meditative pause architecture.
Recommended use	Meditative content, elegy, contemplative narration.

HATTIE — spirited storyteller elder `choir.eld.female.hattie · VOX-08-31`

Character	Front-porch raconteur; mischief intact at eighty.
Parameters	F0 180 Hz (138–240) · FSc 1.12 · Tilt neutral-bright for band · Br 0.24 · Rg 0.18 · Tempo 3.8 · Dyn med-high (2.9 st)
Prosody signature	Dramatic build-and-release; conspiratorial low asides; relishing lengthened key words.
Recommended use	Folk tales, comic elders, family-saga narration.

HESPERA — sinister elder villainess `choir.eld.female.villain.hespera · VOX-08-32`

Character	Malevolent experience; the witch at the end of every wrong road.
Parameters	F0 162 Hz (120–215) · FSc 1.09 · Tilt dark with whistling top edge · Br 0.35 · Rg 0.24 · Tempo 3.2 · Dyn low (1.7 st) with serpentine glides
Prosody signature	Slow rise-fall coils; sweetness curdling mid-phrase; breath made audible as presence.
Recommended use	Witches, curses, gothic horror narration.

9. Villain Voice Design Requirements (VOX-V)

The eight villain voices — **Wick, Briar, Corvin, Sable, Malvern, Ravenna, Grimshaw, Hespera** (profiles VOX-08-04, -08, -12, -16, -20, -24, -28, -32; exactly one per age band × gender cell, per VOX-G-001) — share an engineered "menace grammar" defined by the following requirements.

> **VOX-V-001** MUST There shall be exactly eight villain voices, one per (age band × gender) cell: Wick, Briar, Corvin, Sable, Malvern, Ravenna, Grimshaw, Hespera.

VOX-V-002 **MUST** Villain menace shall be realized through prosodic and spectral design — reduced pitch dynamism with strategic spikes, unnaturally measured tempo, micro-pauses before semantically loaded words, darkened or hollowed spectral balance, and controlled roughness/breathiness — and never through added distortion effects, pitch-shifting artifacts, or "monster filter" post-processing.

Rationale: the villains must be beautiful too; menace is a performance style, not an audio defect.

VOX-V-003 **MUST** Each villain voice shall remain fully capable of neutral and warm delivery when emotion parameters are set accordingly (a villain reading a shopping list neutrally must sound like a person, not a gimmick), so that antagonist characters can feign friendliness in narrative use.

VOX-V-004 **SHOULD** Each villain voice shall expose a documented `menace` emphasis behaviour: at high emotion-intensity settings, the §12 emotion system shall bias toward that voice's signature menace prosody rather than generic excitement.

VOX-V-005 **MUST** The child villain voices (Wick, Briar) shall achieve unease exclusively through prosodic control (over-precision, flattened melody, measured tempo) while retaining fully child-typical F0 and formant structure, and shall comply with VOX-G-010.

Part III — Functional Requirements

10. Text Processing & Linguistic Front End (TXT)

10.1 Input Acceptance

TXT-001 MUST The engine shall accept arbitrary Unicode (UTF-8) input strings from 1 character to at least 1,000,000 characters per request, subject only to documented memory-based limits per platform.

TXT-002 MUST The engine shall never crash, hang, or produce unbounded output on malformed, adversarial, or degenerate input (emoji floods, zero-width characters, control characters, single repeated character × 10^6, mixed scripts). Unsupported characters shall be skipped or spoken via documented fallback rules.

TXT-003 MUST Input shall be accepted as (a) plain text, (b) plain text with SSML-C inline markup (§10.5), or (c) a pre-phonemized sequence (§10.6), selected explicitly by API — never by guessing.

10.2 Text Normalization

TXT-010 MUST The front end shall expand, with correct spoken forms, at minimum: cardinal and ordinal numbers (to at least 10^{15}), decimals, fractions, percentages, currency (£, $, €, with major/minor units), dates in common formats, clock times (12/24-hour), phone-number-style digit strings, years (incl. "1990s"), Roman numerals in title context ("Henry VIII"), units (km, kg, MHz, etc.), common abbreviations (Dr., St., etc. — context-disambiguated: "St. John" vs "Baker St."), URLs and email addresses (spoken structurally), and hyphenated ranges ("3–5" → "three to five").

TXT-011 MUST Normalization shall handle **Scripture reference formats** as a first-class category: "John 3:16" → "John three, sixteen" (style configurable: *chapter-verse*, *chapter colon verse*, or *spoken-full* "John chapter three, verse sixteen"), including ranges ("Rom. 8:28–30"), book abbreviations for all 66 Protestant canonical books, and multi-reference lists.

Rationale: THE ONE is a primary consumer; verse references must never be spoken as "john three colon sixteen."

TXT-012 SHOULD Normalization behaviour shall be configurable via a `NormalizationPolicy` (e.g., verbatim mode for code/identifiers, natural mode for prose, scripture style selection, year style).

TXT-013 SHOULD The engine shall correctly handle common typography: smart quotes, em/en dashes (rendered as appropriate pauses), ellipses (trailing-off prosody), parentheses (subordinated prosody), and ALL-CAPS words (emphasis, configurable).

10.3 Grapheme-to-Phoneme (G2P)

TXT-020 MUST The front end shall include a built-in pronunciation lexicon of at least 120,000 English word forms with primary/secondary stress, plus a trained neural or rule-hybrid G2P fallback for out-of-vocabulary words achieving ≥ 92% phoneme accuracy on a held-out OOV test set.

TXT-021 MUST The G2P system shall correctly disambiguate common heteronyms by part-of-speech context ("read", "lead", "tear", "bass", "bow", "wind", "live", "wound", "minute", "row", at minimum a documented list of ≥ 60 heteronyms).

TXT-022 MUST A **user lexicon** API shall allow consuming apps to register custom pronunciations (word → phonemes or word → respelling) at runtime, persisting per app, taking precedence over the built-in lexicon.

Rationale: theological names (Socinus, Crellius, Athanasius, Melchizedek), game invented names, and brand names must be pronounceable on day one without a package update.

TXT-023 SHOULD The built-in lexicon shall include a curated **biblical & theological proper-noun supplement** of at least 2,500 entries (all canonical personal/place names plus common patristic, Reformation, and Socinian-tradition names).

TXT-024 MUST The phoneme inventory shall be documented (ARPAbet or IPA mapping table published in DOC-004), stable across versions, and exposed publicly for the pre-phonemized input path.

10.4 Sentence Segmentation & Phrasing

TXT-030 MUST The front end shall segment input into sentences and intonational phrases, robust to abbreviations ("Dr. Smith arrived."), decimal points, quotations, and dialogue punctuation, and shall pass phrase-boundary strength (minor/major/paragraph) to the prosody system.

TXT-031 MUST Long sentences (> ~30 words) shall be automatically divided into natural breath groups at syntactically plausible boundaries so that no synthesized breath group exceeds a voice-appropriate maximum duration.

TXT-032 SHOULD Paragraph breaks shall produce longer pauses and a pitch reset appropriate to the voice's pause style; chapter/section breaks (double blank line or explicit markup) longer still.

10.5 SSML-C Markup Dialect

TXT-040 MUST The engine shall parse an inline XML-like markup dialect ("SSML-C") supporting at minimum: <break time="500ms"/> and <break strength="weak|medium|strong|paragraph"/>; <emphasis level="reduced|moderate|strong">; <prosody pitch="±Nst" rate="N%" volume="±NdB"> (nestable); <phoneme ph="…"> (inline pronunciation override); <say-as interpret-as="characters|number|ordinal|date|scripture">; <voice id="…"> (mid-text voice switching); and <mark name="…"/> (timing marker reported in synthesis metadata).

TXT-041 MUST Malformed markup shall degrade gracefully: unparseable tags are stripped and their text content spoken; a diagnostics list of markup warnings shall be returned with the result, and strict mode (throwing on malformed markup) shall be available.

TXT-042 MUST The <voice> tag shall permit multi-character dialogue in a single synthesis request, with per-segment voice/parameter application and no audible discontinuity artifacts at switch points beyond the natural pause.

10.6 Pre-Phonemized & Advanced Input

TXT-050 SHOULD The API shall accept a fully specified phoneme+prosody sequence (phonemes, durations optional, per-phoneme pitch targets optional), bypassing the front end, for research, precise lip-sync, and singing-adjacent experiments.

TXT-051 MAY The front end may expose its intermediate representation (normalized text, phonemes, predicted phrase structure) for inspection/debugging without performing synthesis.

11. Synthesis Core (SYN)

SYN-001 MUST The synthesis pipeline shall be: input → front end (§10) → prosody prediction → acoustic model (mel + durations + F0 + energy) → vocoder → PCM float32 audio at 48 kHz (engine-native), with 44.1 kHz and 24 kHz resampled outputs available (§14).

SYN-002 MUST Synthesis shall be fully deterministic when the caller supplies a seed: identical (text, voice, parameters, seed, package version) shall produce bit-identical audio on the same device class.

Rationale: reproducible video/audiobook builds and testability.

SYN-003 MUST Without a seed, controlled prosodic variation shall be applied so that the same sentence synthesized twice differs naturally (micro-timing, contour), avoiding robotic sameness in repeated game lines.

SYN-004 MUST The engine shall support both operating modes with identical audio quality: **batch** (render full request to buffer/file) and **streaming** (§13).

SYN-005 MUST Every synthesis result shall include metadata: total duration; per-word and per-phoneme timing (start/end ms); sentence boundaries; positions of all `<mark/>` tags; the effective parameter set used; and any diagnostics.

Rationale: verse-highlighting in THE ONE, subtitle/caption sync, lip-sync in games, and timeline placement in video all require timing data — this is a core deliverable, not an extra.

SYN-006 MUST Concurrent synthesis: at least 2 independent synthesis jobs on iOS/macOS (target 4 on M-series) shall run simultaneously without audio corruption, with fair scheduling and per-job cancellation.

SYN-007 MUST Cancellation shall be prompt (< 100 ms to stop compute) and clean (no leaked buffers, model left in reusable state).

SYN-008 MUST The engine shall be crash-isolated from consuming app logic: any internal failure surfaces as a typed error (§19), never a trap, for all inputs satisfying TXT-001/002.

SYN-009 SHOULD A **warm-up API** shall allow preloading models and JIT-compiling compute graphs ahead of first use, so first-synthesis latency can be paid at app launch rather than at the user's first tap.

SYN-010 SHOULD The engine shall expose a lightweight **duration estimate** API (predicted speech duration for text+voice+rate without full synthesis, ±10% accuracy) for layout, pagination, and video planning.

12. Prosody, Emotion & Expression Control (PRO)

12.1 Core Continuous Controls

PRO-001 MUST **Pitch shift**: continuous ±6 semitones from the voice's base, artifact-free across the range, applied in the acoustic model (not post-hoc resampling), preserving formants.

PRO-002 MUST **Rate**: continuous 0.6×–2.0× with duration-model-aware time scaling (pauses and phone durations scale non-uniformly, as natural fast/slow speech does — not linear time-stretch).

PRO-003 MUST **Emotion intensity**: continuous 0.0 (flat, reserved) to 1.0 (fully animated), modulating pitch dynamism, energy contour, tempo micro-variation, and pause expressiveness according to each voice's character (§8 signatures, §9 for villains).

PRO-004 MUST **Breathiness**: continuous 0.0–1.0 around the voice's default, including audible-breath insertion at breath-group boundaries when ≥ threshold (toggleable independently).

PRO-005 MUST **Volume/energy**: continuous gain and dynamic-range (compression) control pre-master, independent of the output loudness normalization of AUD-020.

PRO-006 MUST Age-shift and gender-shift modifiers per VOX-P-007.

12.2 Emotion Styles

PRO-010 MUST Beyond intensity, the engine shall support at least eight discrete emotion styles applicable to any voice: *neutral, joyful, tender, solemn, fearful, angry, sorrowful, suspenseful* — each realized as a coherent prosodic-and-timbral configuration (not merely a pitch offset), blendable with intensity 0–1.

PRO-011 SHOULD Styles shall be applicable at sub-sentence granularity via SSML-C (e.g., `<style name="suspenseful" intensity="0.7">`), with smooth transitions across style boundaries.

PRO-012 MAY A per-voice style-compatibility matrix may down-weight combinations that break character (e.g., *joyful* at 1.0 on Grimshaw is clamped and a diagnostic emitted).

12.3 Fine Control

PRO-020 MUST Word-level emphasis via SSML-C shall produce genuinely accented realization (pitch excursion, lengthening, energy) matching the voice's emphasis style.

PRO-021 MUST Explicit pauses (SSML-C break) shall be sample-accurate to ±10 ms and shall interact correctly with automatic phrasing (no doubled pauses).

PRO-022 SHOULD Question, exclamation, and quotation intonation shall be automatically and voice-appropriately realized from punctuation, with an API switch to suppress automatic interpretation.

PRO-023 MAY A pitch-contour override API (target semitone offsets at anchor points across a phrase) may be provided for expressive fine-tuning and quasi-musical delivery.

12.9 The Customization Envelope

PRO-030 MUST For every voice, the documentation shall publish the guaranteed envelope (min/max for every continuous control within which VOX-G-007 quality holds); outside the envelope the engine shall clamp and report, never emit degraded audio silently.

13. Streaming & Real-Time Operation (STR)

STR-001 MUST Streaming synthesis shall deliver audio incrementally (chunked PCM via `AsyncSequence` and/or callback) with time-to-first-audio per PRF-010, playing while later text is still being synthesized.

STR-002 MUST Chunk boundaries shall be seamless: zero clicks, gaps, or level jumps at chunk joins (verified by null-test against batch render of the same seeded request).

STR-003 MUST Streaming shall sustain RTF < 0.5 on reference devices (§28.1) so playback never underruns after start; if the engine detects it cannot sustain real-time on the current device/thermal state, it shall report this before audible failure and offer degrade options (lower-cost vocoder path per ML-V-006).

STR-004 MUST Timing metadata (SYN-005) shall be delivered incrementally alongside chunks, enabling live word-highlighting and lip-sync during streaming.

STR-005 MUST Mid-stream cancellation and mid-stream parameter changes at sentence boundaries (e.g., game raises emotion intensity for the next line) shall be supported.

STR-006 SHOULD A **dialogue queue** convenience layer shall accept an ordered list of (text, voice, params) utterances and stream them gaplessly or with specified gaps — the primitive for game conversations and multi-voice scenes.

STR-007 SHOULD Streaming output shall be directly attachable to an `AVAudioEngine` source node supplied by the package, so consuming apps get playback with one call while retaining access to raw chunks for custom pipelines.

14. Audio Output, Formats & Export (AUD)

14.1 Native Output

AUD-001 MUST Native engine output: mono float32 PCM, 48 kHz. API accessors shall provide `AVAudioPCMBuffer`, `Data` (interleaved), and `[Float]` forms without copying where feasible.

AUD-002 MUST Built-in sample-rate conversion to 44.1 kHz, 24 kHz, and 16 kHz, and bit-depth conversion to int16, with high-quality resampling (≥ 100 dB stopband).

14.2 File Export

AUD-010 MUST Export encoders: WAV (PCM 16/24-bit), CAF, AAC in .m4a (64–256 kbps), and ALAC in .m4a. MP3 export MAY be provided via the OS encoder where available.

AUD-011 MUST Exported files shall carry correct metadata (duration, sample rate) and optional caller-supplied tags (title, artist/voice, chapter marks for m4a/m4b audiobook export).

AUD-012 MUST **Audiobook export**: multi-chapter m4b assembly from a list of (chapter title, text) pairs, with chapter markers, resumable long-running export, progress reporting, and integrity verification of the final file.

Rationale: Ponte and the theological corpus need one-call book-to-audiobook.

14.3 Mastering & Consistency

AUD-020 MUST Loudness normalization to a caller-selectable target (default −16 LUFS mono for spoken word; −14 and −19 presets available), with true-peak limiting at −1.0 dBTP; all 32 voices shall land within ±0.5 LU of target on standard passages so voices can be intermixed without level jumps.

AUD-021 MUST DC offset ≤ −60 dBFS; no denormal-related CPU spikes; start/end of every rendered utterance fades over ≤ 5 ms to guarantee clicklessness.

AUD-022 SHOULD Optional gentle mastering chain (high-pass at 40 Hz, de-esser, soft-knee limiter) — off by default, on by preset for "broadcast" export.

14.4 Playback Convenience

AUD-030 MUST One-call playback (`engine.speak(…)`) with pause/resume/stop/seek and completion callbacks, cooperating with the app's `AVAudioSession` (never hijacking category/activation on iOS — configuration hooks provided, decisions left to the app).

AUD-031 SHOULD Word-synchronized playback observation: an `AsyncStream` of (word range, timestamp) events during playback for karaoke-style highlighting.

14.7 Video & Timeline Support

AUD-040 SHOULD Export of per-line audio files plus a machine-readable timing sidecar (JSON: file, duration, word timings) suitable for scripted assembly in video editors; and SRT/VTT caption generation from synthesis timing.

15. Caching, Storage & Asset Management (CCH)

CCH-001 MUST A content-addressed synthesis cache shall store rendered audio keyed by hash(text, voice, parameters, seed, engine version); repeat requests shall return cached audio without recomputation, transparently.

CCH-002 **MUST** Cache controls: per-app size cap (default 500 MB, configurable), LRU eviction, pinning (never-evict for app-critical audio, e.g., THE ONE's rendered Gospels), full and selective purge, and usage inspection API.

CCH-003 **MUST** Cache storage shall respect platform conventions (Caches directory, excluded from backup by default; pinned items optionally in Application Support with backup) and shall survive package version updates when the engine version component of the key is unchanged.

CCH-010 **MUST** Model assets shall support two distribution modes selectable by the consuming app: (a) fully embedded in the app bundle; (b) On-Demand Resources / background-downloaded asset packs with integrity verification (SHA-256), resumable download, and per-voice-family granularity — with mode (a) always available for the sovereignty guarantee (§30 SEC-001 offline mandate).

CCH-011 **MUST** Total asset budget per PRF-030; asset loading shall be lazy (voice families load on first use or explicit warm-up) and unloadable under memory pressure with automatic reload.

CCH-012 **SHOULD** Background batch rendering: a queue API that renders large jobs opportunistically (charging, idle) using `BGProcessingTask` on iOS, with persistence across app restarts.

Part IV — The Swift Package & API

16. Package Structure & Distribution Form (PKG)

PKG-001 MUST CHOIR shall be delivered as a Swift Package (SwiftPM) importable with a single `import CHOIR`, buildable in Xcode with no external package-manager steps (no CocoaPods, no Carthage, no pip at integration time).

PKG-002 MUST Module layout: `CHOIR` (public API), `CHOIRCore` (engine internals, non-public), `CHOIRAssets` (model resources or asset-pack loader), and optional `CHOIRSpatial` (visionOS spatial layer) — with the public surface confined to `CHOIR`.

PKG-003 MUST Minimum deployment targets: iOS 17, iPadOS 17, macOS 14, visionOS 1.0, watchOS 10; Swift 5.9+ with full Swift 6 strict-concurrency compliance (no data-race warnings under complete checking).

PKG-004 MUST Zero third-party runtime dependencies. Only Apple frameworks (Foundation, Accelerate, CoreML, AVFoundation, etc.) may be linked by the runtime package.

PKG-005 MUST All training-pipeline code, licences, and datasets shall be licence-clean for closed-source commercial distribution of the runtime (no GPL/AGPL contamination of shipped assets or code; permissive-licensed research code use documented in a NOTICE file).

PKG-006 SHOULD The package shall also be exportable as a signed XCFramework for integration contexts where source distribution is undesirable.

PKG-007 MUST App Store compliance: no private API usage; all model execution via public Core ML/Accelerate; package passes App Store review requirements as embedded in a shipping app.

17. Public API Requirements (API)

17.1 API Design Principles

API-001 MUST The API shall offer three concentric tiers, each complete for its use level: **Tier 1 (one-liner):** `try await Choir.speak("text", voice: .maeve)` — synthesize and play. **Tier 2 (standard):** engine instance, `SynthesisRequest` with parameters, results with audio + metadata, export calls. **Tier 3 (expert):** streaming, dialogue queues, phoneme input, contour overrides, cache and asset management.

API-002 MUST All potentially long operations shall be `async`; synchronous variants only where trivially fast (metadata queries, duration estimates from cache). All public types shall be `Sendable` where semantically possible.

API-003 MUST Voices shall be addressable both type-safely (`Voice.grimshaw`, autocomplete-discoverable, grouped by age/gender namespaces) and dynamically by stable string ID (`Voice(id:)`) for data-driven use (game character tables).

API-004 MUST `SynthesisParameters` shall be a value type with a documented default per voice, builder-style modification, Codable persistence, and validation that reports envelope clamping (PRO-030).

API-005 MUST The full public API shall carry DocC documentation: every public symbol documented, with articles and runnable snippets per DOC section; no undocumented public symbol may ship.

API-006 MUST API stability: semantic versioning; no source-breaking change within a major version; deprecations marked one minor version before behavioural change.

API-007 SHOULD A SwiftUI convenience layer shall ship: `SpeakButton`, a voice-picker component, and a word-highlighting text view driven by AUD-031 events — usable but never required.

API-008 MAY An AppIntents/Shortcuts surface may expose synthesis to the Shortcuts app from consuming apps that opt in.

17.2 Representative Signatures (normative shape, not final names)

```
let engine = try await ChoirEngine()                        // loads lazily, honours warm-up
let result = try await engine.synthesize(
    .init(text: "In the beginning was the word of God.",
          voice: .alaric,
          parameters: .init(rate: 0.95, emotion: .solemn(0.6))))
result.audio          // AVAudioPCMBuffer
result.timing.words   // [(range, start, end)]
try await result.export(to: url, format: .m4a(bitrate: 128))

for try await chunk in engine.stream(request) { play(chunk) }   // streaming tier
```

18. Concurrency, Threading & Cancellation (CON)

CON-001 MUST No public API call shall ever block the main thread for more than 1 ms; all compute occurs on engine-owned executors/queues.

CON-002 MUST Structured-concurrency native: all async APIs honour Swift task cancellation (checked at least every 50 ms of compute) and propagate it per SYN-007.

CON-003 MUST The engine shall be an actor-isolated or internally synchronized object safe to call from any thread/task without external locking; multiple engine instances in one process shall be supported (with shared, reference-counted model weights).

CON-004 MUST QoS discipline: interactive requests (streaming for UI/games) run at `.userInitiated`; background batch at `.utility` / `.background`; priorities shall be caller-assignable per request and honoured by the internal scheduler.

CON-005 MUST Under OS memory-pressure notifications the engine shall shed caches and unloadable assets before the app is jetsam-killed, and shall restore transparently afterwards.

19. Error Model & Reliability (REL)

REL-001 MUST A single public `ChoirError` taxonomy shall cover: input errors (empty, oversize, invalid markup in strict mode), asset errors (missing voice pack, corrupt asset, insufficient storage), resource errors (memory, thermal refusal), execution errors (model failure), and cancellation — each with a stable code, localized description, and recovery suggestion.

REL-002 MUST Partial-failure policy for batch/multi-item jobs shall be caller-selectable: fail-fast, or continue-and-report (result carries per-item success/error list).

REL-003 MUST The engine shall run a start-up self-check (asset hashes, model load, 0.5 s silent smoke synthesis) exposed as `engine.verify()`, so consuming apps can detect a broken installation deterministically rather than failing at first user request.

REL-004 MUST Mean crashes attributable to CHOIR in any consuming app: zero known reproducible crashes at release; fuzzing per TST-006 is the gate.

REL-005 SHOULD Structured logging via `os.Logger` subsystems (off/error/info/debug levels, caller-configurable), never logging user text content above debug level (privacy, SEC-004).

Part V — Platforms

20. Cross-Platform Requirements (PLT)

PLT-001 MUST The identical public API shall compile and behave identically across iOS, iPadOS, macOS, and visionOS; watchOS exposes a documented subset (§24). Platform divergence shall exist only where hardware demands it, and every divergence shall be documented in a single "Platform Differences" DocC article.

PLT-002 MUST Audio output produced from the same seeded request shall be perceptually identical across platforms (allowing only numerical differences below −60 dB null-test residual between Apple-silicon devices).

PLT-003 MUST The engine shall run on Apple silicon (A14 and later, all M-series) as the supported hardware baseline; Intel Macs MAY be supported in CPU-only degraded-performance mode or explicitly excluded — whichever is chosen shall be documented and enforced at package resolution.

21. iOS / iPadOS Requirements

PLT-010 MUST Full feature set on iPhone and iPad, including streaming, batch, background rendering (CCH-012), and audiobook export, within iOS background-execution rules.

PLT-011 MUST Correct behaviour across audio interruptions (calls, Siri), route changes (headphones, CarPlay, AirPods), and backgrounding: playback state machine per AUD-030 survives and resumes per app policy.

PLT-012 SHOULD Thermal adaptation: on sustained load the engine monitors `ProcessInfo.thermalState` and steps down to the efficient vocoder path (ML-V-006) before the OS throttles, reporting the transition.

22. macOS Requirements

PLT-020 MUST Full feature set plus workstation-scale batch: parallel rendering of at least 4 concurrent jobs on M-series, CLI-friendly operation (the engine callable from a command-line tool target for scripted audiobook/video builds), and no App-Sandbox violations when sandboxed.

PLT-021 SHOULD A shipped `choir-cli` tool (render text file → audio, batch manifest → audiobook) for the developer's own pipelines.

23. visionOS & Spatial Audio Requirements (SPA)

SPA-001 MUST On visionOS (and iOS/macOS where AVAudioEnvironmentNode applies), CHOIR shall provide a spatial playback layer: any utterance or dialogue-queue voice attachable to a 3D position/orientation, with distance attenuation and reverb-blend parameters, via a `SpatialSpeaker` abstraction wrapping AVAudioEngine spatialization (and PHASE where beneficial).

SPA-002 MUST Multiple simultaneous spatial speakers (≥ 6 concurrent positioned voices on Vision Pro) shall mix correctly with head-tracked rendering, each independently controllable (position updates ≥ 30 Hz without artifacts).

SPA-003 SHOULD Convenience scene bindings: attach a `SpatialSpeaker` to a RealityKit `Entity` so a character entity speaks from wherever it is, position tracked automatically.

SPA-004 MAY Ambisonic/bed export (render a scene's dialogue to a spatial audio file) may be provided for video production.

SPA-005 MUST The spatial layer shall be an optional module (`CHOIRSpatial`): apps that ignore it carry no RealityKit/PHASE linkage cost.

24. watchOS & tvOS Requirements

PLT-030 MUST watchOS shall support a documented subset: a curated on-watch voice set (≥ 4 voices within the watch asset budget of PRF-032), short-utterance synthesis (≤ 60 s per request), core parameters (pitch/rate/emotion), and playback — with the same API shapes, returning a typed capability error for unsupported features.

PLT-031 SHOULD A phone-assist mode: the watch app may request rendering from the paired iPhone via WatchConnectivity for full-library access, transparently through the same API when available.

PLT-032 MAY tvOS support (full library, streaming) may be added; if trivially attainable from the shared codebase it SHOULD be enabled.

Part VI — Models & Machine Learning

This part constrains the training pipeline only insofar as its outputs (deployed models) must satisfy runtime requirements. Architecture choices remain free where unconstrained.

25. Acoustic Model Requirements (ML-A)

ML-A-001 MUST A single multi-voice acoustic model shall map (phoneme sequence, prosodic features, voice embedding, control parameters) → (mel-spectrogram, durations, F0 contour, energy), non-autoregressive or chunk-parallel so that generation speed satisfies PRF RTF targets.

ML-A-002 MUST All §12 continuous controls shall be model-native conditioning inputs (trained, not post-processing), guaranteeing artifact-free behaviour across the published envelopes.

ML-A-003 MUST The 32 voice embeddings shall occupy a structured, documented latent space in which the §7 age/gender trajectories are approximately linear directions, enabling VOX-P-007 modifiers and (optionally) VOX-G-009 interpolation.

ML-A-004 MUST Training data shall consist only of (a) speech data whose licence permits closed-source commercial synthesis use, and/or (b) synthetically generated or parametrically designed speech, with full provenance recorded per PKG-005; no scraped, unlicensed, or personally identifying voice data.

ML-A-005 MUST The complete training pipeline shall be reproducible: pinned dependency versions, fixed seeds where framework-supported, configuration-as-code, and a single documented command per stage; retraining from scratch shall reproduce models passing all quality gates.

ML-A-006 SHOULD The model shall support duration/pitch teacher-forcing inputs to serve TXT-050 pre-phonemized precision mode.

26. Vocoder Requirements (ML-V)

ML-V-001 MUST The vocoder shall convert mel frames to 48 kHz PCM with quality per §29 (no metallic ringing, no phase artifacts, no buzz on sustained vowels, clean fricatives and plosives) across all 32 voices including high-roughness villain timbres.

ML-V-002 MUST The vocoder shall be causal or low-lookahead (≤ 100 ms) to satisfy streaming TTFA targets.

ML-V-003 MUST Designed roughness/breathiness (jitter, shimmer, subharmonics, aspiration noise) shall be faithfully reproduced as intended texture — the vocoder must not "clean up" the elderly and villain voices.

ML-V-006 MUST Two operating points shall ship: **Quality** (default; meets §29 fully) and **Efficient** (≥ 2× faster, watch/thermal fallback; MOS within 0.4 of Quality), switchable per request and automatically under PLT-012/STR-003 with reporting.

27. Core ML Conversion, Quantization & Compute Units (ML-C)

ML-C-001 MUST All runtime networks shall ship as Core ML packages (mlprogram), executing via ANE and/or GPU where the operator set allows, CPU fallback always functional.

ML-C-002 MUST Quantization (e.g., FP16 baseline; INT8/palettization where quality-safe) shall be applied per-network to meet the PRF-030 asset budget, with a documented quality-delta report proving each quantized network passes §29 gates (null-test and MOS delta vs FP32 reference).

ML-C-003 MUST Conversion shall be scripted and reproducible (ML-A-005 applies); every shipped .mlmodelc shall be traceable to a training run ID and conversion config hash recorded in the asset manifest.

ML-C-004 SHOULD Compute-unit selection shall be automatic with per-device profiles (prefer ANE on A-series for battery; GPU on M-series for throughput), overridable by API for testing.

Part VII — Quality Attributes

28. Performance Requirements (PRF)

28.1 Reference Devices

All numeric targets are measured on: **R1** iPhone 13 (A15) — the floor mobile device; **R2** current-generation iPhone Pro; **R3** MacBook (M2 or later); **R4** Apple Watch Series 9; **R5** Apple Vision Pro. Targets are for the Quality vocoder path unless stated.

PRF-001 MUST Batch RTF ≤ 0.35 on R1, ≤ 0.15 on R3 (i.e., a 60-minute audiobook renders in ≤ 9 minutes on a Mac).

PRF-010 MUST Streaming TTFA (warm engine): ≤ 350 ms on R1, ≤ 250 ms on R2/R3/R5 for a typical sentence; sustained streaming RTF ≤ 0.5 on R1.

PRF-011 MUST Cold start (process launch → first audio, including model load): ≤ 3.0 s on R1, ≤ 1.5 s on R3; warm-up API (SYN-009) reduces first-request TTFA to the PRF-010 numbers.

PRF-020 MUST Peak memory during single-stream synthesis: ≤ 350 MB on R1, ≤ 180 MB on R4 (Efficient path), measured RSS attributable to CHOIR.

PRF-021 MUST Battery: 60 minutes of continuous streaming synthesis+playback on R1 shall consume ≤ 12% battery; background batch shall respect Low Power Mode by pausing or degrading per policy.

PRF-030 MUST Total shipped asset size (all 32 voices, both vocoder paths, lexicon): ≤ 400 MB installed; ≤ 250 MB SHOULD be targeted. Per-voice-family asset packs shall each be ≤ 120 MB for ODR delivery.

PRF-032 MUST watchOS asset subset ≤ 60 MB installed.

PRF-040 MUST A reproducible benchmark harness (part of the repo) shall measure every PRF number on every reference device per release; releases ship with the benchmark report.

29. Audio Quality Requirements (QUA)

QUA-001 MUST Structured listening MOS (developer-administered, blinded A/B protocol defined in TST-010) shall average ≥ 4.2/5 for naturalness-as-designed and ≥ 4.5/5 for audio cleanliness across a 20-passage standard corpus per voice.

QUA-002 MUST Zero audible artifacts on the standard corpus: no clicks, buzzes, metallic ringing, warbling, phase swirl, dropped/duplicated phones, or gibberish insertions. Any single artifact instance on the corpus is a release blocker.

QUA-004 MUST Intelligibility: word-level transcription accuracy ≥ 98% when standard Harvard sentences synthesized by each voice (default parameters) are transcribed by an independent ASR system, and ≥ 96% at envelope corners (VOX-G-007).

QUA-005 MUST Long-form stability: a continuous 60-minute render shall exhibit no drift in loudness (±0.5 LU), tempo (±3%), or timbre (spectral centroid ±5%) from first to last 5 minutes.

QUA-006 MUST Correct pronunciation on the theological/biblical proper-noun supplement (TXT-023): 100% on a 300-word audit list.

QUA-007 SHOULD Listener fatigue: 30-minute continuous listening self-test per narration-tagged voice with no fatigue-inducing quality noted (sibilance harshness, pumping, monotony) — logged as a structured checklist.

QUA-008 MUST §8 profile conformance: each shipped voice measured within tolerance of its specified parameters — median F0 ±10%, tempo ±8%, and characteristic descriptors confirmed in the blinded identification test of VOX-G-003 (this is §29.6 referenced by §8).

30. Privacy, Security & Ethical Requirements (SEC)

SEC-001 MUST The runtime shall make **zero network connections**, ever, for any purpose (no telemetry, no licence checks, no asset fetches except the explicit, caller-initiated ODR mode of CCH-010). Verified by network-audit test in TST.

SEC-002 MUST No analytics, no identifiers, no persistent records of synthesized text beyond the local cache (CCH), which the consuming app fully controls and can purge.

SEC-003 MUST Asset integrity: model assets hash-verified at load; corrupt assets produce a typed error, never undefined behaviour.

SEC-004 MUST Logging shall never emit user text at default levels (REL-005); debug-level text logging shall be opt-in and documented.

SEC-005 MUST The package shall contain no capability for cloning real voices (restating VOX-G-002 as a runtime property: no speaker-encoder input path exists in shipped binaries).

SEC-006 SHOULD Documentation shall include a responsible-use note for consuming apps (synthetic-voice disclosure norms for published media).

31. Accessibility Requirements (ACC)

ACC-001 MUST CHOIR playback shall coexist correctly with VoiceOver and system speech (audio ducking behaviour configurable; never fighting the screen reader for the audio session).

ACC-002 SHOULD The word-timing stream (AUD-031) shall be documented as the basis for accessible karaoke highlighting, and caption export (AUD-040) as the basis for deaf/hard-of-hearing parity in consuming apps.

ACC-003 MAY An AVSpeechSynthesisProviderAudioUnit implementation may be provided so CHOIR voices appear as system voices selectable in iOS/macOS accessibility settings.

32. Language & Localization Requirements (LOC)

LOC-001 MUST v1.0 language scope: English (General American and Received Pronunciation British variants selectable per request; default per-voice documented). All 32 voices support both variants.

LOC-002 MUST The architecture (phoneme inventory abstraction, front-end modularity, conditioning design) shall be documented as multi-language-ready so that future languages (priorities: Latin ecclesiastical pronunciation, Dutch, German, Hungarian — the Ponte corpus languages) are additive work, not redesign.

LOC-003 SHOULD Foreign words and proper names embedded in English text (Latin titles, Greek transliterations, Hebrew names) shall be speakable via lexicon entries and the `<phoneme>` tag with acceptable anglicized realization.

LOC-004 MAY Error strings and API-facing localizable text may ship localized; the package's own strings shall at minimum be localization-ready (String Catalogs).

Part VIII — Integration & Lifecycle

33. Integration Requirements (INT)

33.1 THE ONE (Bible study app)

INT-001 MUST Verse-addressed synthesis: given (translation, book, chapter, verse-range, voice), CHOIR-side support shall make it a single call for THE ONE to obtain audio with **per-verse timing boundaries** in metadata, enabling verse highlighting and tap-to-seek.

INT-002 MUST Scripture normalization style (TXT-011) shall be selectable app-wide; divine-name and archaic-form pronunciations (e.g., "LORD", "thee/thou/-eth" forms in KJV/YLT) shall be correct out of the box.

INT-003 SHOULD Chapter pre-render + pinning (CCH-002) workflow documented as a recipe: background-render a book with a chosen voice (Maeve and Alaric are the anticipated defaults) and pin for offline listening.

INT-004 SHOULD Corpus-work narration: long-document rendering of translated Socinian works (the app's corpus) via the audiobook path (AUD-012) with footnote-skipping markup conventions documented.

33.2 Ponte (PDF translation)

INT-010 MUST Long-document robustness: a 300-page translated document (≈ 150k words) shall render to audiobook in one resumable job with progress, pause/resume across app restarts, and per-section files or a single chaptered m4b.

INT-011 SHOULD Structural markup mapping documented: headings → chapter marks + pause/reset, block quotes → subtle delivery shift, footnotes → skip or defer policies.

33.3 Games

INT-020 MUST The dialogue-queue (STR-006) + low-latency streaming (PRF-010) + spatial layer (§23) together shall support: dynamic (non-prerecorded) NPC lines at conversation pace, interruptible barks, per-character voice+parameter presets loadable from data files (API-003), and deterministic seeds for repeatable cutscenes.

INT-021 SHOULD Phoneme timing (SYN-005) shall be documented with a viseme mapping table for lip-sync in RealityKit/SceneKit/Unity-via-plugin contexts (mapping table is deliverable; Unity plugin is out of scope v1.0).

33.4 Video Production

INT-030 MUST Script-to-takes workflow: a manifest of lines (character → voice/params, text) renders to per-line files + timing sidecar + SRT (AUD-040), deterministic under seeds, suitable for the Midjourney→Kling→CapCut style pipeline.

INT-031 MAY Multiple takes per line (seed variations) may be batch-generated with a take-selection manifest format.

33.5 Showcase App (package-facing constraints only)

INT-040 MUST Everything the showcase app displays (gallery demos VOX-G-008, live parameter control, export, spatial demo) shall be achievable through the public API alone — the showcase uses no internal hooks, proving the API's completeness.

34. Documentation Requirements (DOC)

DOC-001 MUST A DocC catalog shall ship in-package with: Getting Started (integration in ≤ 10 minutes), the three API tiers with runnable examples, per-platform notes (PLT-001), and one recipe article per §33 integration scenario.

DOC-002 MUST A **Voice Book**: one page per voice — character description, §8 parameters, envelope (PRO-030), demo passages, and recommended pairings (e.g., "Maeve + solemn 0.4 for Psalms").

DOC-003 MUST SSML-C reference: every tag, attribute, edge case, and a validated example corpus.

DOC-004 MUST Phoneme inventory table (TXT-024) with audio examples per phoneme.

DOC-005 MUST Maintenance manual: full training-pipeline runbook (ML-A-005), asset build/conversion steps (ML-C-003), benchmark procedure (PRF-040), and release checklist — written for the developer's future self with zero assumed context.

DOC-006 SHOULD A one-page cheat sheet (printable) of the Tier-1/Tier-2 API and all voice IDs.

35. Versioning, Distribution & Maintenance (DST)

DST-001 MUST Semantic versioning of the package; a separate **engine-version** integer governs cache keys (CCH-001) and bumps only when audio output changes for identical requests.

DST-002 MUST Voice stability: within a major version, a given (voice, request, seed, engine-version) renders identically forever; audible voice changes require a major version and migration notes.

DST-003 MUST The repository shall contain everything required to rebuild the product from scratch — runtime source, training pipeline, asset build scripts, benchmark harness, test suites, documentation source — under version control with tagged releases.

DST-004 SHOULD Additive evolution paths documented for: new voices, new languages (LOC-002), new emotion styles — each without breaking existing identifiers or caches.

DST-005 MUST Yearly-OS-release resilience: no reliance on deprecated APIs at ship time; a documented smoke-test procedure for each new OS beta.

36. Testing & Acceptance Requirements (TST)

TST-001 MUST Unit coverage ≥ 85% of front-end logic (normalization, G2P dispatch, segmentation, SSML-C parsing) with golden-file tests for every TXT-010 category and every TXT-011 scripture form.

TST-002 MUST Golden-audio regression: a corpus of ≥ 200 seeded requests across all voices whose rendered audio is hash/null-test compared per release; any unexplained diff blocks release (DST-002 enforcement).

TST-003 MUST Full QUA gate execution per release candidate (MOS protocol, artifact audit, intelligibility ASR run, long-form stability render), results archived.

TST-004 MUST PRF benchmark run on all reference devices per release (PRF-040).

TST-005 MUST Platform integration tests: interruption/route-change matrix on iOS (PLT-011), sandbox compliance on macOS, watch subset behaviour, and a visionOS spatial smoke scene.

TST-006 MUST Fuzzing: ≥ 24 machine-hours of randomized/malformed input against the front end and API per release with zero crashes/hangs (REL-004 gate); corpus of found cases becomes permanent regression tests.

TST-007 MUST Network-silence audit: automated verification that the runtime opens no sockets in embedded-asset mode (SEC-001).

TST-008 SHOULD Concurrency stress: 4-job parallel synthesis soak (2 h) with cancellation storms, memory-pressure injection, and thermal simulation.

TST-010 MUST The MOS listening protocol shall be written down (passage list, blinding method, scoring rubric, fatigue limits) before first scoring, and reused unchanged across releases for comparability.

37. Requirements Index & Traceability Summary

Area	Prefix	Count	Highest-risk requirements
Voice library general	VOX-G	10	VOX-G-003 (distinctness), VOX-G-006 (single shared model)
Acoustic design params	VOX-P	7	VOX-P-007 (runtime age/gender shift)
Voice profiles	VOX-08	32	All villain profiles; elderly texture (design vs artifact line)
Villain design	VOX-V	5	VOX-V-002 (menace without effects)
Text front end	TXT	19	TXT-011 (scripture), TXT-020 (G2P), TXT-042 (voice switching)
Synthesis core	SYN	10	SYN-002 (determinism), SYN-005 (timing metadata)
Prosody & emotion	PRO	13	PRO-003/010 (emotion system), PRO-030 (envelopes)
Streaming	STR	7	STR-002 (seamless chunks), STR-003 (sustain guarantee)
Audio out/export	AUD	13	AUD-012 (audiobook), AUD-020 (loudness parity)
Cache & assets	CCH	7	CCH-010 (dual asset modes)
Package	PKG	7	PKG-004 (zero deps), PKG-005 (licence purity)
Public API	API	8	API-001 (tiering), API-006 (stability)
Concurrency	CON	5	CON-003 (thread-safe engine)
Errors/reliability	REL	5	REL-004 (zero-crash gate)
Platforms	PLT	10	PLT-002 (cross-device parity), PLT-030 (watch subset)
Spatial	SPA	5	SPA-002 (6 concurrent spatial voices)
ML acoustic	ML-A	6	ML-A-004 (data licensing), ML-A-005 (reproducibility)
ML vocoder	ML-V	4	ML-V-003 (texture fidelity), ML-V-006 (dual path)
Core ML	ML-C	4	ML-C-002 (quantization vs quality)
Performance	PRF	9	PRF-010 (TTFA), PRF-030 (asset budget)
Quality	QUA	8	QUA-001/002 (MOS & zero-artifact gates)
Privacy/ethics	SEC	6	SEC-001 (network silence), SEC-005 (no cloning path)
Accessibility	ACC	3	ACC-001 (VoiceOver coexistence)
Localization	LOC	4	LOC-002 (multi-language readiness)
Integration	INT	11	INT-001 (verse timing), INT-020 (game latency stack)

Documentation	DOC	6	DOC-005 (maintenance manual)
Distribution	DST	5	DST-002 (voice permanence)
Testing	TST	10	TST-002 (golden audio), TST-006 (fuzzing)

Total: 239 numbered requirements (plus the 32 binding voice-profile specifications of §8 counted as one requirement each within VOX-08).

> **Reading order for implementation planning.** Parts VI and VII contain the highest technical risk (model quality vs. on-device budget); Part II defines the product's soul; Part IV defines its daily usability. The recommended next documents, per the agreed preproduction sequence, are the Architecture Document and the Voice Design Spec — both of which now have a complete requirements skeleton to trace against.

End of CHOIR-SRS-001 v1.0. All requirements subject to revision only by versioned amendment of this document.